

Estimating Ancestry Composition from GWAS Summary Statistics

Allele-frequency and LD-moment channels: derivations,
estimators, and diagnostics

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ppb - Cross-ancestry PGS portability benchmark

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1 Summary

This report derives and compares methods for estimating the ancestry composition of a GWAS summary-statistics dataset from reference data alone. Two families of estimators are considered. The first regresses the per-variant effect-allele frequencies (EAF) reported in the summary statistics on per-population reference frequencies; this channel is implemented in the SMARTpred service and follows Summix [3, 11] and the bigsnpr ancestry resources [7, 10]. The second family, which is the focus of this report, uses no allele frequencies at all: it extracts the ancestry composition from the *second moments of the association statistics* — products $z_i z_j$ and squared statistics χ_i^2 — against per-ancestry LD reference panels. This is the channel pioneered by DISTMIX [6] and ZMIX [5], and it is grounded in the same moment identity that underlies LD-score regression [4].

The practical motivation is threefold. First, some summary statistics ship without a usable frequency column, or with one that cannot be trusted; the LD channel needs only $\hat{\beta}_i$ and SE_i . Second, the two channels are statistically independent summaries — the first moment of the frequencies versus the second moment of the effects — so their agreement is itself a strong data-quality diagnostic, in the spirit of the misspecification checks of Privé et al. [9]. Third, for cross-ancestry polygenic-score work such as the portability measurements of this repository, an estimate of the ancestry composition of every input dataset is a prerequisite for asserting that a “target ancestry” dataset is what it claims to be.

The report is organised as follows. Section 2 fixes notation. Section 3 states the frequency-based estimator for reference. Sections 4 and 5 derive what the association statistics know about the analysed population’s LD. Sections 6 and 7 define the two LD-moment estimators. Sections 8 – 10 treat uncertainty, identifiability, and failure modes; Section 11 positions the estimators against prior work; Section 12 maps them onto the SMARTpred and ppb code bases.

2 Setting and notation

Let $i = 1, \dots, m$ index variants (in practice a HapMap 3 panel [2] of roughly one million variants, harmonised across references), and let K reference ancestries be available with LD correlation matrices $R^{(k)} = (\rho_{ij}^{(k)})$ on the standardized-genotype scale, estimated from ancestry-matched reference panels (for example 1000 Genomes super-populations [1]; in the ldpred3 ecosystem, UK Biobank references of the LDpred2 family [8]). The analysed GWAS dataset provides

$$z_i = \frac{\hat{\beta}_i}{\text{SE}_i}, \quad \chi_i^2 = z_i^2, \quad (1)$$

with effective sample size n (per-variant n_i where available), and optionally a reported effect-allele frequency f_i . The unknown ancestry composition is

$$\pi = (\pi_1, \dots, \pi_K) \in \Delta_K = \left\{ \pi : \pi_k \geq 0, \sum_k \pi_k = 1 \right\}. \quad (2)$$

3 The frequency channel (implemented)

If the analysed population is a mixture of the reference populations with little drift since admixture, its allele frequencies satisfy to first order

$$f_i \approx \sum_{k=1}^K \pi_k p_i^{(k)} + \varepsilon_i, \quad (3)$$

with $p_i^{(k)}$ the panel frequency of variant i 's counted allele in ancestry k . The estimator minimises the residual sum of squares over the simplex,

$$\hat{\pi} = \arg \min_{\pi \in \Delta_K} \sum_{i=1}^m \left(f_i - \sum_k \pi_k p_i^{(k)} \right)^2, \quad (4)$$

after matching variants by identifier and allele pair (swapped alleles flip f_i ; strand-ambiguous variants are excluded, since frequency-based strand resolution is unreliable exactly where ancestries differ). This is the Summix estimator [3] up to optimisation details, and it is what the SMARTpred ancestry mode implements with chromosome jackknife standard errors and panel-confusability diagnostics. Its requirement — a trustworthy EAF column — is precisely what the channels of Sections 6–7 dispense with.

4 What the association statistics know

Write the marginal association statistic of variant i as

$$z_i = \frac{1}{\sqrt{n}} x_i^\top y, \quad y = \sum_{j=1}^m x_j \gamma_j + \varepsilon, \quad (5)$$

where x is the standardized genotype matrix of the *analysed* population A (mean zero, variance one per variant), the per-variant true standardized effects satisfy $\text{Var}(\gamma_j) = h^2/m$ (infinitesimal model), $\text{Var}(\varepsilon) = 1 - h^2$, and n is the effective sample size. Taking expectations over effects and noise, with $R^{(A)} = (\rho_{ij}^{(A)})$ the LD of the analysed population,

$$\mathbb{E}[z_i z_j] = \frac{nh^2}{m} (R^{(A)})_{ij} + (1 - h^2) \rho_{ij}^{(A)} \quad \text{for all } i, j, \quad (6)$$

and, setting $i = j$ with $\rho_{ii}^{(A)} = 1$ and $\ell_i^{(A)} = \sum_j (\rho_{ij}^{(A)})^2$ the LD score of variant i in the analysed population,

$$\mathbb{E}[\chi_i^2] = 1 - h^2 + \frac{nh^2}{m} \ell_i^{(A)} \approx a + b \ell_i^{(A)}, \quad (7)$$

the LD-score-regression identity [4]: the intercept a absorbs $1 - h^2$ plus any confounding inflation, and the slope $b = nh^2/m$ is the polygenic signal. Equations (6)–(7) state that the first two moments of the association statistics are proportional to the LD structure of the population that was analysed. That structure — not the effect sizes — is what identifies the ancestry.

Two quantitative remarks. First, for a pair of variants in LD, both terms of (6) scale with the analysed population's own correlation; for unlinked pairs both terms vanish, so unlinked pairs carry no ancestry information. Second, the noise-covariance term dominates the tagging term by roughly

$$\frac{(1 - h^2)}{(nh^2/m) \bar{\ell} \cdot \rho} \longleftrightarrow \frac{1 - h^2}{nh^2 \bar{\ell}/m} \approx \frac{0.75}{0.05} \quad \text{at } n = 10^5, h^2 = 0.25, m = 10^6, \bar{\ell} \approx 2 : \quad (8)$$

the pair-product moment is driven by correlated estimation noise, which is present for *any* trait with correct standard errors, including ones with negligible polygenic signal. The χ^2 moment, by contrast, requires genuine signal ($h^2 > 0$), since at $h^2 = 0$ the expectation in (7) is flat in ℓ .

5 Mixture model for the analysed population's LD

Assume the analysed population is a random mating of the K source populations and that, for the variant separations used below, the haplotype segment spanning a pair carries a single ancestry, drawn with probability π_k (admixture-LD is deferred to Section 10). Then the

standardized-genotype correlation in A is, to first order, the ancestry-weighted mixture

$$\rho_{ij}^{(A)} \approx \sum_{k=1}^K \pi_k \rho_{ij}^{(k)} \quad \text{for linked } (i, j). \quad (9)$$

Substituting (9) into the dominant term of (6) keeps the pair moment *linear in* π . The χ^2 route is different: the LD score squares the correlation, so

$$\ell_i^{(A)} = \sum_j \left(\sum_k \pi_k \rho_{ij}^{(k)} \right)^2 = \sum_k \sum_{k'} \pi_k \pi_{k'} \ell_i^{(kk')}, \quad \ell_i^{(kk')} := \sum_j \rho_{ij}^{(k)} \rho_{ij}^{(k')}, \quad (10)$$

which is linear in the *products* $\pi_k \pi_{k'}$, not in π itself. The bilinear scores $\ell_i^{(kk')}$ — computable once from each ordered pair of reference panels — are the design variables of Section 7. Note that regressing χ^2 on the plain per-ancestry scores $\ell_i^{(k)} = \ell_i^{(kk)}$ instead is misspecified whenever the cross-ancestry correlation vectors overlap, which they always do; the cross terms $\pi_k \pi_{k'} \sum_j \rho_{ij}^{(k)} \rho_{ij}^{(k')}$ are exactly what that shortcut drops.

6 Estimator A: pair products (linear in π)

For a set P of sampled variant pairs with non-trivial reference LD, use the dominant term of (6)–(9) with an unconstrained scale $s > 0$ absorbing $(1 - h^2)$ and the tagging term:

$$\mathbb{E}[z_i z_j] \approx s \sum_{k=1}^K \pi_k \rho_{ij}^{(k)}, \quad (i, j) \in P. \quad (11)$$

Estimate the composition by constrained least squares over the simplex, profiling out the scale:

$$(\hat{\pi}, \hat{s}) = \arg \min_{\pi \in \Delta_K, s \geq 0} \sum_{(i,j) \in P} \left(z_i z_j - s \sum_k \pi_k \rho_{ij}^{(k)} \right)^2. \quad (12)$$

For fixed π , the optimal s is the least-squares ratio $\langle zz, L\pi \rangle / \|L\pi\|^2$ with L the pair-by-ancestry design matrix, so (12) reduces to a one-dimensional search over a simplex-constrained least-squares problem — the same numerical problem, in fact, that the frequency channel solves, and solvable by a pure-NumPy active-set method. Requirements and properties:

1. **No EAF column, no h^2 required.** The dominant noise term exists for any trait with correctly calibrated standard errors (Section 4); what it needs is adequate n and honest SEs.
2. **Pairs must be linked.** Unlinked pairs have $\rho_{ij}^{(k)} \approx 0$ for every k and contribute only noise; take pairs within LD blocks, e.g. all pairs with $\max_k |\rho_{ij}^{(k)}|$ above a floor, capped per block to bound cost.
3. **Genome-wide z 's.** Because each pair is weakly informative, the estimator needs statistics across all chromosomes; a single region does not suffice.

This is the estimator family of ZMIX [5]; DISTMIX [6] uses the same information indirectly, scoring each ancestry by how well its LD imputes the observed z -pattern.

7 Estimator B: bilinear LD-score regression

Regress the LDSC-style truncated statistic on the $K(K+1)/2$ bilinear score vectors of (10):

$$\mathbb{E}[\min(\chi_i^2, c)] = a + \sum_{k \leq k'} b_{kk'} \ell_i^{(kk')}, \quad c \approx 30, \quad (13)$$

with non-negativity constraints $b_{kk'} \geq 0$ (they equal $s \pi_k \pi_{k'}$ at truth) and an unconstrained intercept. Under (7)–(10) the population values are

$$b_{kk'} = \frac{nh^2}{m} \pi_k \pi_{k'}, \quad (14)$$

so the composition is recovered from the row sums of the estimated coefficient matrix, normalised:

$$\hat{\pi}_k = \frac{\sum_{k'} \hat{b}_{kk'}}{\sum_k \sum_{k'} \hat{b}_{kk'}}. \quad (15)$$

The overall scale nh^2/m cancels; $\hat{\pi}$ is then projected onto Δ_K if needed, and the projection distance is itself a specification diagnostic, since at truth $\hat{B} \approx \hat{s} \hat{\pi} \hat{\pi}^\top$ is a scaled rank-one matrix. Per-variant sample sizes enter multiplicatively in (14): either restrict to variants with comparable n_i or fit with signal term ℓ_i/n_i . Estimator B needs genuine polygenic signal (flat χ^2 at $h^2 = 0$ carries no LD profile) and well-powered statistics, but it is a single per-variant regression over m rows — cheaper than the pair sampler of Estimator A once the bilinear scores are precomputed.

8 Uncertainty

Residuals of both estimators are correlated across nearby variants and pairs through LD, so i.i.d. standard-error formulas are wrong. Use the delete-one-chromosome (or delete-one-LD-block) jackknife: refit on all but one chromosome’s variants (pairs), and form

$$\text{SE}(\hat{\pi}_k) = \sqrt{\frac{g-1}{g} \sum_{g'=1}^g (\hat{\pi}_k^{(-g')} - \bar{\hat{\pi}}_k)^2}, \quad (16)$$

with g informative groups; the same construction the frequency channel uses. For Estimator A, $\text{Var}(z_i z_j) \approx 1 + \{\mathbb{E}[z_i z_j]\}^2$ under approximate normality suggests weights $1/(1 + \hat{\rho}_{\text{fit},ij}^2)$; the unweighted fit with jackknife is the simpler default. For Estimator B the LDSC truncation already tames the worst heteroskedasticity.

9 Identifiability

In all three channels only the *relative* proportions are identified: the frequency channel is exact only up to the noise scale, and the LD channels carry an unknown slope (s , or nh^2/m) that is not separable from π without external knowledge of h^2 . This is the same target as Summix and ZMIX — a composition vector, not an heritability estimate. The deeper identifiability constraint is geometric: the ancestry design columns (frequencies $p^{(k)}$, pair correlations $\rho_{ij}^{(k)}$, or bilinear scores $\ell^{(kk')}$) can be severely collinear across continents. LD patterns in particular decay with distance similarly everywhere, so the LD design is worse conditioned than the frequency design; the pairwise-column correlations and the Gram condition number should be reported alongside any estimate (the confusability diagnostics already implemented for the frequency panel generalise verbatim), and near-collinear ancestries should be pooled rather than reported separately.

10 Assumptions and failure modes

Assumption 1 (Random mating, segment-wise LD). *The mixture approximation (9) holds for linked pairs. Admixture-LD violates it: the admixture process creates long-range covariance of the form $\pi_k(1 - \pi_k) d_i^{(k)} d_j^{(k)}$, with $d_i^{(k)}$ the ancestry-informative frequency contrast, inflating pair products at ancestry-informative markers and biasing $\hat{\pi}$. The bias decays with admixture age; recent admixture is the hard case.*

Assumption 2 (Infinitesimal effects). $\text{Var}(\gamma_j) = h^2/m$ with exchangeable effects on the admixed-genotype scale. A trait driven by a few large loci violates the moment structure of (6)–(14); the truncation in (13) mitigates this for Estimator B, less so for Estimator A.

Assumption 3 (Honest standard errors). *The dominant term of Estimator A is the correlated-noise covariance: miscalibrated, winsorised, or meta-analysis-heterogeneity-distorted SEs contaminate it directly. This is the channel’s greatest sensitivity.*

Assumption 4 (No confounding beyond the intercept). *Stratification and uncontrolled confounding add LD-score-shaped and pair-product-shaped components (the LDSC intercept story of Bulik-Sullivan et al. [4]). Free intercepts absorb only the mean part; $\hat{\pi}$ drifts toward the LD profile of the confounding structure.*

Assumption 5 (Harmonised references). *All K references are built on the same variant panel with consistent allele orientation, so pair correlations and bilinear scores are comparable across ancestries. Panel construction from 1000 Genomes [1] or UK Biobank references [8] must pass through the same harmonisation as the target statistics.*

11 Relation to prior work

Table 1 summarises the channels. *Summix* [3] and its successor [11] deconvolve ancestry from allele frequencies with a mixture model, and adjust frequencies to a target composition; the bigsnpr ancestry resources [7, 10] implement the same idea through projection onto reference PCs followed by simplex-constrained quadratic programming, with a worldwide 21-group reference. *DISTMIX* [6] infers mixing proportions in order to build ancestry-informed LD matrices for imputing summary statistics; *ZMIX* [5] regresses z -score products on per-population genotype correlations by constrained least squares — Estimator A of Section 6 — and reports, in simulation, frequency-based methods near-perfect while the z -channel runs a few percentage points off, worse for admixed targets: association statistics simply carry less ancestry information than frequencies. The misspecification checks of Privé et al. [9] are the binary special case of all of this — “does the reference match the statistics?” — where the estimators here answer “in what proportions?”. Estimator B’s bilinear scores are, to the best of our knowledge, new as an ancestry estimator; as a construction they are the multi-ancestry generalisation of LD-score regression [4], and nothing in this report claims priority over methods we have not surveyed exhaustively.

Table 1: Ancestry-composition channels from summary statistics.

Channel	Data used	Moment	Prior work
Frequency	EAF column f_i	first, (3)	Summix [3], [11], bigsnpr [10]
Pair products	$z_i z_j$, linked pairs	second, (11)	ZMIX [5], DISTMIX [6]
Bilinear LD scores	χ_i^2 , all variants	second, (13)	LDSC generalisation [4]; new here

12 Implementation mapping and recommendation

In the SMARTpred service, the frequency channel is implemented (`smartpred/ancestry.py`) with hash-pinned 1000 Genomes panels, chromosome jackknife standard errors, and confusability reporting. Of the LD channels, we recommend Estimator A as the second channel to implement: it needs no EAF column and no h^2 , it reduces to the same simplex-constrained least-squares solver already in place, and the per-ancestry LD references and their LD-score sidecars already exist in the service’s registry. The pair set P should be built from the registered references’ blocks (pairs within blocks with $\max_k |\rho_{ij}^{(k)}|$ above a floor, capped per block), with delete-one-chromosome jackknife standard errors (16). Estimator B is one further step once bilinear scores $\ell^{(kk')}$ are precomputed per reference pair, and is the natural benchmark arm against Estimator A in a simulation study before either is exposed on the public service; a block-wise Gaussian likelihood on the full z -covariance remains research-grade (per-block cubic cost, plus every misspecification of Assumptions 1–4 at once).

For this repository, the estimators slot into the cross-ancestry portability pipeline as a pre-flight check: estimate the composition of every “target ancestry” dataset, verify it against the dataset’s declared ancestry, and refuse or flag measurements whose target is a mixture the reference cannot separate — the quantitative generalisation of the reference-matching requirement stated by Privé et al. [9].

Acknowledgements

The derivations in Sections 4–7 were worked out in discussion of the SMARTpred ancestry mode; the frequency channel’s implementation there served as the numerical template. All citations were verified against the published records; any error in them is ours.

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